CHAPTER 4

HUMAN HEALTH AND DISEASE

CHAPTER ESSENCE AT A GLANCE:
Innate and acquired immune cascades, pathogenic life cycles, oncogenic transformation, and retroviral pathogenesis.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Common Infectious Pathogens and Clinical Manifestations:** Bacterial: Typhoid (Salmonella typhi, confirmed by Widal test, sustained fever, intestinal perforation) and Pneumonia (Streptococcus pneumoniae, fluid-filled alveoli). Viral: Common cold (Rhinovirus, infects nasal passages but spares lungs). Protozoan: Malaria (Plasmodium vivax and malignant P. falciparum; female Anopheles vector injects sporozoites; liver schizogony and erythrocyte rupture releases hemozoin toxin causing periodic chills and fever) and Amoebiasis (Entamoeba histolytica). Helminthic: Filariasis (Wuchereria bancrofti, Culex vector, elephantiasis). Fungal: Ringworm (Microsporum, Trichophyton).

• **Innate Barriers and Acquired Humoral Cellular Immunity:** Innate immunity (non-specific, present from birth): Physical (skin, mucus), Physiological (acid in stomach, lysozyme in tears), Cellular (PMNL neutrophils, monocytes, macrophages), and Cytokine (interferons secreted by virus-infected cells protecting uninfected cells). Acquired immunity (specific, memory): Humoral immunity mediated by B-lymphocytes producing antibodies (H2L2 monomer: two heavy and two light chains joined by disulfide bonds; IgG crosses placenta, IgA in colostrum, IgE in allergies); Cell-Mediated Immunity (CMI) mediated by T-lymphocytes responsible for graft rejection during organ transplants.

• **Active Passive Immunization, Allergies, and Autoimmunity:** Active immunity is slow but long-lasting, generated by host antibody production during infection or vaccination. Passive immunity provides pre-formed antibodies (anti-tetanus serum, anti-venom, colostrum IgA). Allergies: IgE-mediated hypersensitivity triggering histamine and serotonin release from mast cells. Autoimmune disease: immune system attacks self-antigens (Rheumatoid arthritis). Lymphoid organs: Primary (bone marrow, thymus for maturation) and Secondary (spleen, lymph nodes, tonsils, Peyer's patches, MALT: 50% of lymphoid tissue).

• **AIDS Pathogenesis and Cancer Neoplastic Transformation:** AIDS: Caused by HIV (retrovirus with ssRNA genome); reverse transcriptase synthesizes viral cDNA, which integrates into host DNA; replicates inside macrophages (HIV factory) and progressively attacks and destroys Helper T-cells (CD4+ T-lymphocytes), depleting counts below 200 per mm^3; diagnosed by ELISA. Cancer: Loss of contact inhibition leading to uncontrolled mitosis; benign versus malignant tumors; metastasis is the migration of sloughed neoplastic cells via bloodstream to establish secondary tumors; proto-oncogenes activated by carcinogens (radiation, chemicals).

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Antibody Architecture} = H_2L_2 \quad [2\ \text{Heavy Chains} + 2\ \text{Light Chains linked by S-S bridges}]$$

Immunoglobulin Monomer Structural Formula

$$\text{HIV Replication} = \text{Viral RNA} \xrightarrow{\text{Reverse Transcriptase}} \text{Viral DNA} \xrightarrow{\text{Integrase}} \text{Host Genome} \rightarrow CD4^{+}\ \text{Lysis}$$

Retroviral Reverse Transcription and Lymphoid Depletion

$$\text{Malarial Toxin} = \text{RBC Rupture} \rightarrow \text{Hemozoin Release} \rightarrow \text{Cyclic Chills and High Fever (48-72h)}$$

Pathogenesis of Periodic Malarial Paroxysms

$$\text{Innate Defense} = \text{Physical (Skin)} + \text{Physiological (HCl)} + \text{Cellular (PMNL)} + \text{Cytokine (Interferons)}$$

Four Primary Lines of Non-Specific Innate Defense

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Graft rejection following kidney or organ transplantation is mediated by CELL-MEDIATED IMMUNITY (CMI) driven by T-lymphocytes, not humoral immunity.

• **Alert:** In HIV infection, macrophages act as the continuous HIV factory, but the virus progressively destroys Helper T-cells (CD4+), collapsing immunity.

• **Alert:** Metastasis is the most feared characteristic of malignant cancerous tumors, in which cells detach and invade distant organs via blood circulation.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What are interferons? Which cells produce them, and how do they protect uninfected neighboring cells? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Interferons: Specialized cytokine proteins secreted by virus-infected cells. **[1 Mark]**

• Mechanism: They diffuse to neighboring uninfected cells and stimulate them to produce antiviral proteins that block viral replication, containing the spread of viral infection. **[1 Mark]**

2. Why must a patient who undergoes kidney transplantation take immunosuppressive drugs for life? Which branch of the immune system is responsible for graft rejection? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Responsible branch: Cell-Mediated Immunity (CMI), mediated by T-lymphocytes that recognize non-self foreign HLA antigens on the transplanted graft. **[1 Mark]**

• Immunosuppressant need: Immunosuppressive drugs (e.g., Cyclosporin A) must be taken for life to continuously suppress T-cell activation and prevent immune destruction (rejection) of the donor organ. **[1 Mark]**

3. Draw a diagram of an antibody molecule. Label the antigen-binding sites, light chains, heavy chains, and disulfide bonds, and explain why it is represented as H2L2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Structure: Each antibody molecule consists of four polypeptide chains: two identical longer heavy (H) chains and two identical shorter light (L) chains bound by inter-chain disulfide bonds. **[1 Mark]**

• Why H2L2: Because it possesses exactly two heavy chains and two light chains in each monomeric immunoglobulin unit. **[1 Mark]**

• Antigen-binding sites: The variable N-terminal regions of each heavy and light chain pair form two identical antigen-binding sites (paratopes) that fit specific epitope shapes. **[1 Mark]**

4. Describe the asexual life cycle of Plasmodium in the human host, explaining the release of hemozoin and its clinical manifestation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Infection and liver stage: Female Anopheles mosquito injects infectious sporozoites into human blood; sporozoites reach the liver and reproduce asexually within hepatocytes, bursting liver cells. **[1 Mark]**

• Erythrocytic stage: Released merozoites infect red blood cells (erythrocytes) and reproduce asexually, causing periodic synchronized rupture of RBCs. **[1 Mark]**

• Hemozoin release: Rupture of RBCs releases a toxic crystalline byproduct of hemoglobin digestion called hemozoin, which triggers recurring chills and high fever every 3 to 4 days. **[1 Mark]**

5. Explain the replication cycle of the Human Immunodeficiency Virus (HIV) inside human macrophages and helper T-cells (CD4+). [Score: 3 Marks]

Answer & Key Valuation Steps:

• Entry and reverse transcription: HIV binds CD4 receptors and enters the host macrophage; its viral RNA genome is reverse transcribed into complementary viral DNA by the enzyme reverse transcriptase. **[1 Mark]**

• Integration: Viral DNA enters the host nucleus and integrates into the host chromosome via integrase, directing the cell to synthesize new viral RNA and proteins; macrophages act as an HIV factory. **[1 Mark]**

• Helper T-cell lysis: HIV simultaneously enters Helper T-lymphocytes (CD4+), replicates, and lyses the cells; progressive depletion of helper T-cells destroys acquired immunity, leading to opportunistic infections. **[1 Mark]**

6. Distinguish between benign and malignant tumors. Explain the phenomenon of metastasis and why it is considered the most feared property of cancer. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Benign tumors: Remain confined to their original site, do not spread to other parts of the body, and cause relatively little damage. **[1 Mark]**

• Malignant tumors: Mass of proliferating neoplastic cells that grow rapidly, invade surrounding healthy tissues, and compete for vital nutrients. **[1 Mark]**

• Metastasis: Sloughed cancer cells detach from primary tumors and travel via blood or lymph to distant organs, establishing secondary malignant colonies; it makes cancer difficult to eradicate surgically. **[1 Mark]**

7. A patient experiences recurring high fever accompanied by severe shivering chills every 48 hours. A peripheral blood smear reveals ring-stage trophozoites within red blood cells. Identify the pathogen, name the vector, explain the biochemical cause of the fever spikes, and state the confirmatory test for typhoid if the patient also exhibits sustained high fever and abdominal pain. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Pathogen identification: Plasmodium vivax (causative protozoan of benign tertian malaria). **[1 Mark]**

• Vector: Female Anopheles mosquito. **[1 Mark]**

• Cause of fever spikes: Synchronized rupture of schizont-infected erythrocytes releases hemozoin toxin and pyrogens into the blood circulation every 48 hours, acting on the hypothalamic thermostat to induce chills and high fever. **[1 Mark]**

• Confirmatory test for typhoid: The Widal serological agglutination test confirms Salmonella typhi infection. **[1 Mark]**